



CNR

CRAFT N ROAM

AI Travel Decision Simulator & 3D Walk

Craft a transparent plan. Roam the winner in 3D.

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Generative AI · Final Project Presentation

The Problem

Trip planning means balancing budget, time, and taste across dozens of options — and every existing tool skips the reasoning.

Listing sites

Google Maps, TripAdvisor, Yelp show everything and decide nothing.

AI chatbots

One suggestion, no visible tradeoffs — a black box.

Either way

No way to get a real feel for a place before you commit.

Craft, Then Roam

A transparent, traveler-adjustable decision engine, followed by a review-grounded 3D preview of the winner.

1. Craft the plan

- Traveler describes their trip in free text (69 tokens)
- LLM parses it into structured weights
- Deterministic engine scores & ranks 5 curated spots
- Sliders let travelers fine-tune live, for free

2. Roam in 3D

- Traveler unlocks all spots in 3D once (89 tokens)
- LLM reads that spot's curated reviews
- Extracts ground / sky / palette / props (fixed catalog)
- Seeded layout builds a walkable Three.js scene

Demo — Craft a Transparent Plan

"A relaxed five-hour trip, mid-range budget, good food"

- Cours Saleya Market scores 0.94 — strong match, perfect cost & time fit
- Sliders adjust budget, time, and category balance
- Re-ranks instantly — no extra tokens, no black box

CNR - Craft N Roam
Craft your trip, get a transparent plan, then roam the top spot in 3D. 231 tokens

[← Start over](#)

"You want a relaxing 5-hour trip with a mid-range budget and good food included."
Cours Saleya Market scored highest (0.94) - strong match on preferences (0.80), cost fit (1.00), and time fit (1.00).
Promenade des Anglais was runner-up (0.93) but scored lower on time fit.

Fine-tune your plan
Adjusting these re-ranks instantly - no extra tokens.

Budget level: 3 / 5

Time available: 5 hrs

Relax: 0.80 Adventure: 0.20

Culture: 0.30 Food: 0.80

Cours Saleya Market 94

Demo — Roam the Winner

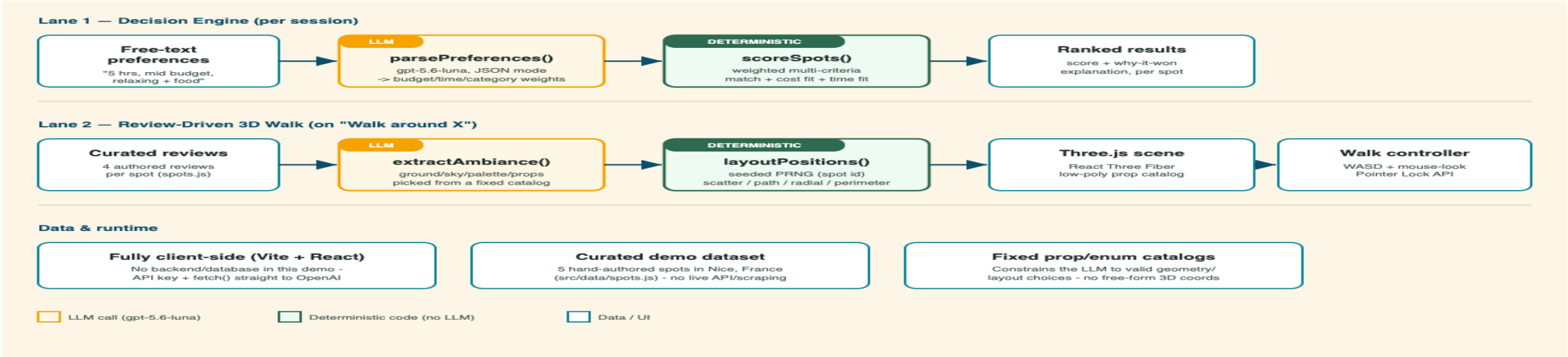
- LLM reads Cours Saleya Market's reviews
- Picks ground, sky, palette, and props from a fixed catalog
- A seeded algorithm places everything — same input, same scene
- Explorable in first person: WASD + mouse-look
- Share 3 photos + a review per spot, earn 30 tokens back



Stylized, not photorealistic — by design (see limitations).

How the AI Is Used

Two narrow LLM extraction steps; the decisions themselves — including the live editor — are ordinary deterministic code.



One Design Principle

The LLM translates. It never decides.

LLM calls

- Free text -> structured weights
- Reviews -> structured ambiance spec
- Fixed JSON schema, fixed enum catalogs
- Validated & clamped before use

Deterministic code

- Weighted multi-criteria scoring
- Live plan-editor re-ranking
- Seeded PRNG for 3D layout
- Same inputs -> same outputs, always

A Real Integration Bug We Hit

gpt-5.6-luna rejects function-tool calling combined with its default reasoning_effort on /v1/chat/completions.

```
invalid_request_error - param: reasoning_effort
```

```
"Function tools with reasoning_effort are not supported for gpt-5.6-luna in /v1/chat/completions."
```

Fix: request raw JSON (response_format: json_object) instead of tool-calling, and pass reasoning_effort: "none" explicitly.

One Honest Limitation

The 3D scenes are stylized, low-poly interpretations — not photorealistic reconstructions of the real locations.

True 3D reconstruction needs dozens of calibrated, overlapping photos (photogrammetry / NeRF) — not review text or a handful of casual photos. We chose to be upfront about this rather than overclaim, and built a deterministic, review-grounded stylized scene instead. The token economy is the same story: presented in full, built as local state only — no real payment happens anywhere in this demo.

Also flagged: WASD movement needs a final real-browser verification pass (automated headless testing was inconclusive due to background-tab throttling).

The Business Model (Presentation Only)

Simulated in the demo as local app state — no real payment processing exists today.

Item	Price
Small token bundle	\$0.80 -> 100 tokens
Large token bundle	\$4.00 -> 600 tokens (~17% off)
Generate a plan (search)	69 tokens (~\$0.46-0.55)
Unlock 3D walking (all spots)	89 tokens (~\$0.60-0.71, one-time)
Share 3 photos + a review, per spot	+30 tokens back

Margin check

Actual LLM cost per search: a fraction of a cent. Priced at 69 tokens (~\$0.50), that is roughly two orders of magnitude of margin before hosting, support, and refunds.

What's Next

Headline idea: crowdsourced photos, real rewards

Travelers upload up to 3 of their own photos per spot -> vision model extracts richer ambiance -> traveler earns a real 30-token reward per spot. Fully legitimate (user-submitted, not scraped). Needs accounts, storage, and moderation — a real backend, deliberately out of scope for now.

- Real payment processing and persistent accounts for the token economy
- Move the OpenAI call server-side before any public deployment
- Hotel, flight, and other token-gated booking add-ons
- Expand beyond Nice to more destinations
- Touch controls for mobile (today assumes a keyboard)
- A real usability study outside the two-person team

Thank you

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<https://github.com/GongNT/travel-3d-simulator>

